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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

Roll Number:2520030011
Name:P.Tejaswini

Generative AI and AI Platforms

25CS2207E

Title

Comparison of BPE, WordPiece and SentencePiece Tokenizers

Aim

To implement and compare BPE, WordPiece and SentencePiece tokenizers using English, Telugu and Hindi text and observe the tokens, token IDs and token counts.

Source Code

```
from transformers import AutoTokenizer

# Pretrained tokenizer models
TOKENIZERS = {
    "BPE - GPT-2": "gpt2",
    "WordPiece - Multilingual BERT": "bert-base-multilingual-cased",
    "SentencePiece - XLM-R": "xlm-roberta-base"
}

# Input sentences in three languages
TEXTS = {
    "English": "Artificial intelligence is changing the world.",
    "Telugu": "కృత్రిమ మేధస్సు ప్రపంచాన్ని మారుస్తోంది.",
    "Hindi": "कृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।"
}


# Load tokenizers
def load_tokenizers():
    loaded = {}

    for display_name, model_name in TOKENIZERS.items():
        print(f"Loading {display_name}...")
        loaded[display_name] = AutoTokenizer.from_pretrained(model_name)

    return loaded


# Compare tokenizers
def compare_tokenizers(loaded_tokenizers):
    for language, text in TEXTS.items():
```

```
        print("\n" + "=" * 75)
        print(f"LANGUAGE: {language}")
        print(f"INPUT TEXT: {text}")
        print("=" * 75)

        for tokenizer_name, tokenizer in loaded_tokenizers.items():

            encoded = tokenizer(
                text,
                add_special_tokens=True,
                truncation=False
            )

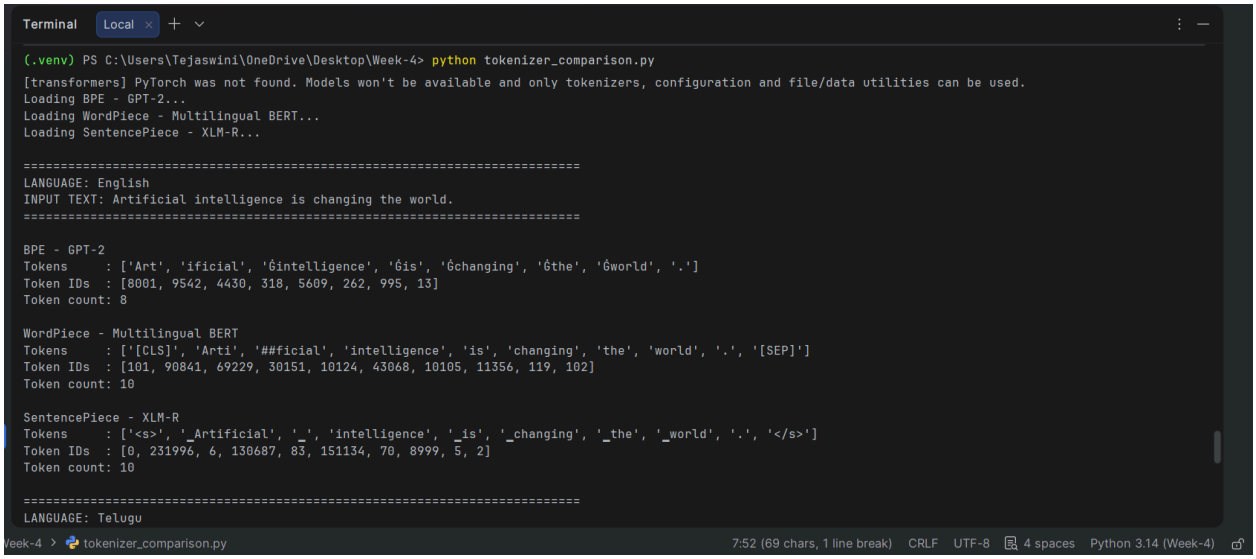
            token_ids = encoded["input_ids"]
            tokens = tokenizer.convert_ids_to_tokens(token_ids)

            print(f"\n{tokenizer_name}")
            print("Tokens     :", tokens)
            print("Token IDs  :", token_ids)
            print("Token count:", len(token_ids))


# Main function
def main():
    tokenizers = load_tokenizers()
    compare_tokenizers(tokenizers)


if __name__ == "__main__":
    main()
```

Output



```
Terminal  Local

(.venv) PS C:\Users\Tejaswini\OneDrive\Desktop\Week-4> python tokenizer_comparison.py
[transformers] PyTorch was not found. Models won't be available and only tokenizers, configuration and file/data utilities can be used.
Loading BPE - GPT-2...
Loading WordPiece - Multilingual BERT...
Loading SentencePiece - XLM-R...

======================================================================
LANGUAGE: English
INPUT TEXT: Artificial intelligence is changing the world.
======================================================================

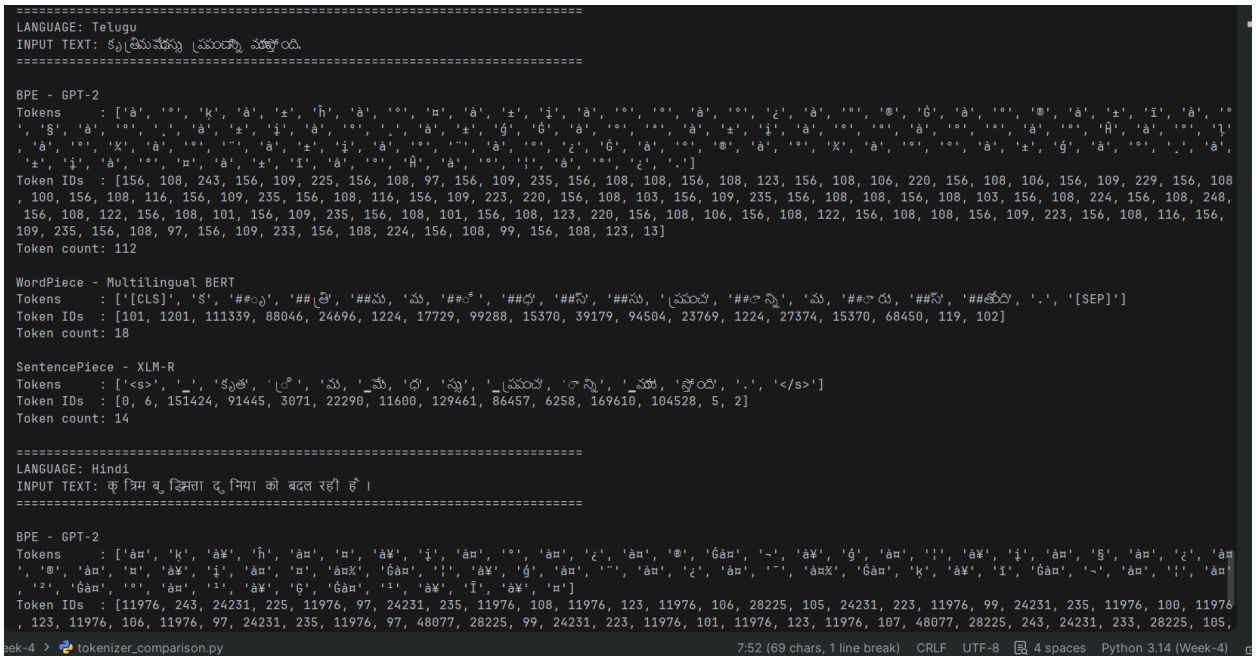
BPE - GPT-2
Tokens     : ['Art', 'ificial', 'Ġintelligence', 'Ġis', 'Ġchanging', 'Ġthe', 'Ġworld', '.']
Token IDs  : [8001, 9542, 4430, 318, 5609, 262, 995, 13]
Token count: 8

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'Arti', '##ficial', 'intelligence', 'is', 'changing', 'the', 'world', '.', '[SEP]']
Token IDs  : [101, 90841, 69229, 30151, 10124, 43068, 10105, 11356, 119, 102]
Token count: 10

SentencePiece - XLM-R
Tokens     : ['<s>', '_Artificial', '_', 'intelligence', '_is', '_changing', '_the', '_world', '.', '</s>']
Token IDs  : [0, 231996, 6, 130687, 83, 151134, 70, 8999, 5, 2]
Token count: 10

======================================================================
LANGUAGE: Telugu
```

```
Week-4 > tokenizer_comparison.py   7:52 (69 chars, 1 line break)  CRLF  UTF-8  4 spaces  Python 3.14 (Week-4)
```



```
======================================================================
LANGUAGE: Telugu
INPUT TEXT: కృత్రిమమేధస్సు ప్రపంచాన్ని మార్చుతోంది.
======================================================================

BPE - GPT-2
Tokens     : ['à', '°', 'Ĺ', 'à', '±', 'Ĥ', 'à', '°', '¤', 'à', '±', 'ĭ', 'à', '°', '°', 'à', '°', '¿', 'à', '°', '®', 'Ĝ', 'à', '°', '®', 'à', '±', 'ī', 'à', '°
', '§', 'à', '°', '.', 'à', '±', 'ĭ', 'à', '°', '.', 'à', '±', 'ģ', 'Ĝ', 'à', '°', '°', 'à', '±', 'ĭ', 'à', '°', '°', 'à', '°', '°', 'à', '°', 'Ĥ', 'à', '°', 'Ļ
', 'à', '°', 'Ĺ', 'à', '°', '¯', 'à', '±', 'ĭ', 'à', '°', '¯', 'à', '°', '¿', 'Ĝ', 'à', '°', '®', 'à', '°', 'Ĺ', 'à', '°', '°', 'à', '±', 'ģ', 'à', '°', '.', 'à',
 '±', 'ĭ', 'à', '°', '¤', 'à', '±', 'Ī', 'à', '°', 'Ĥ', 'à', '°', '¦', 'à', '°', '¿', '.']
Token IDs  : [156, 108, 243, 156, 109, 225, 156, 108, 97, 156, 109, 235, 156, 108, 108, 156, 108, 123, 156, 108, 106, 220, 156, 108, 106, 156, 109, 229, 156, 108
, 100, 156, 108, 116, 156, 109, 235, 156, 108, 116, 156, 109, 223, 220, 156, 108, 103, 156, 109, 235, 156, 108, 108, 156, 108, 103, 156, 108, 224, 156, 108, 248,
 156, 108, 122, 156, 108, 101, 156, 109, 235, 156, 108, 101, 156, 108, 123, 220, 156, 108, 106, 156, 108, 122, 156, 108, 108, 156, 109, 223, 156, 108, 116, 156,
109, 235, 156, 108, 97, 156, 109, 233, 156, 108, 224, 156, 108, 99, 156, 108, 123, 13]
Token count: 112

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'క', '##ృ', '##త్ర', '##మ', 'మ', '##ే', '##ధ', '##స్', '##ను', 'ప్రపంచ', '##ాన్ని', 'మ', '##ార', '##్చ', '##తోంది', '.', '[SEP]']
Token IDs  : [101, 1201, 111339, 88046, 24696, 1224, 17729, 99288, 15370, 39179, 94504, 23769, 1224, 27374, 15370, 68450, 119, 102]
Token count: 18

SentencePiece - XLM-R
Tokens     : ['<s>', '_', 'కృత', 'ి', 'మ', '_మే', 'ధ', 'స్సు', '_ప్రపంచ', 'ాన్ని', '_మార్చు', 'తోంది', '.', '</s>']
Token IDs  : [0, 6, 151424, 91445, 3071, 22290, 11600, 129461, 86457, 6258, 169610, 104528, 5, 2]
Token count: 14

======================================================================
LANGUAGE: Hindi
INPUT TEXT: कृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।
======================================================================

BPE - GPT-2
Tokens     : ['à¤', 'Ĺ', 'à¥', 'Ĥ', 'à¤', '¤', 'à¥', 'į', 'à¤', '°', 'à¤', '¿', 'à¤', '®', 'Ġà¤', '¬', 'à¥', 'ģ', 'à¤', '¦', 'à¥', 'į', 'à¤', '§', 'à¤', '¿', 'à¤
', '®', 'à¤', '¤', 'à¥', 'į', 'à¤', '¤', 'à¤¾', 'Ġà¤', '¦', 'à¥', 'ģ', 'à¤', '¨', 'à¤', '¿', 'à¤', '¯', 'à¤', '¾', 'Ġà¤', 'Ĺ', 'à¥', 'Ī', 'Ġà¤', '¬', 'à¤', '¦', 'à¤
', '²', 'Ġà¤', '°', 'à¤', '¹', 'à¥', 'Ģ', 'Ġà¤', '¹', 'à¥', 'Ī', 'à¥', '¤']
Token IDs  : [11976, 243, 24231, 225, 11976, 97, 24231, 235, 11976, 108, 11976, 123, 11976, 106, 28225, 105, 24231, 223, 11976, 99, 24231, 235, 11976, 100, 11976
, 123, 11976, 106, 11976, 97, 24231, 235, 11976, 97, 48077, 28225, 99, 24231, 223, 11976, 101, 11976, 123, 11976, 107, 48077, 28225, 243, 24231, 233, 28225, 105,
 11976, 99, 11976, 110, 28225, 108, 11976, 117, 24231, 222, 28225, 117, 24231, 230, 24231, 97]
Token count: 68

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'क', '##ृ', '##त्', '##रि', 'ब', '##ुद्ध', '##िम', '##त्ता', 'दुनिया', 'को', 'ब', '##दल', 'रही', 'है', '।', '[SEP]']
Token IDs  : [101, 865, 37500, 18321, 50419, 887, 48010, 50419, 94034, 83874, 11267, 887, 99989, 57203, 10569, 920, 102]
Token count: 17

SentencePiece - XLM-R
Tokens     : ['<s>', '_कृत', 'ि', 'म', '_बुद्धि', 'म', 'त्ता', '_दुनिया', '_को', '_बदल', '_रही', '_है', '।', '</s>']
Token IDs  : [0, 74707, 97885, 1920, 105202, 1920, 54297, 29405, 629, 14512, 8906, 460, 125, 2]
Token count: 14

(.venv) PS C:\Users\Tejaswini\OneDrive\Desktop\Week-4>
```